Emergent Individuality and Neural Integration in Whole-Brain Connectome Simulations of Drosophila melanogaster

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Abstract

We present the first embodied whole-brain spiking simulation of Drosophila melanogaster integrating Hebbian synaptic plasticity and multi-theory neural integration proxy metrics. Using the complete FlyWire v783 connectome (138,639 neurons, 15,091,983 directed weighted synapses), we instantiate two identical spiking neural networks as leaky integrate-and-fire (LIF) neurons on GPU, each driving an independent biomechanical fly body through the NeuroMechFly v2 framework in MuJoCo physics. Both flies share the same initial connectome but receive independent sensory input through compound eyes (721 ommatidia per eye), bilateral olfactory receptor neurons, tarsal gustatory receptors, and Johnston's organ mechanosensory neurons. Over extended simulation (>100 seconds simulated time across 8 paired sessions), Hebbian plasticity (eta = 1e-4, alpha = 1e-7) modifies all synaptic weights independently in each brain. We measure neural integration using proxies derived from four theories: Integrated Information Theory (Phi), Global Workspace Theory (broadcast), Self-Model Theory (sensorimotor correlation), and Perturbation Complexity (cascade analysis). Results show that: (1) both flies converge to a stable composite integration index attractor (CI ~ 0.22 vs 0.20, 12% asymmetry); (2) behavioral profiles diverge dramatically (81% vs 47% escape behavior); (3) 76,045 synapses (0.50%) develop measurable inter-individual divergence; and (4) cross-individual CI correlation is weak (r = 0.190), indicating near-independent neural dynamics. These findings demonstrate that the connectome architecture of Drosophila, combined with embodied sensory experience and synaptic plasticity, is sufficient to generate computational individuality from identical initial conditions. All code and data are publicly available at https://github.com/erojasoficial-byte/fly-brain.

Keywords: connectome, whole-brain simulation, Drosophila, spiking neural network, Hebbian plasticity, neural integration, individuality, embodied cognition

1. Introduction

The adult Drosophila melanogaster brain contains approximately 139,000 neurons connected by over 50 million synapses, making it the most complex nervous system for which a complete synaptic-resolution connectome has been reconstructed (Dorkenwald et al., 2024; Schlegel et al., 2024). The FlyWire project mapped every neuron and synapse in a single female fly brain using electron microscopy, providing an unprecedented substrate for computational neuroscience.

Previous computational work has exploited this connectome in two largely separate lines of research. Shiu et al. (2024) demonstrated that a leaky integrate-and-fire (LIF) model of the central brain accurately predicts sensorimotor circuit responses, but this model lacks a body and operates in open-loop stimulation. Independently, the NeuroMechFly v2 framework (Lobato-Rios et al., 2024) provides a detailed biomechanical fly body in MuJoCo physics with sensory systems, but uses simplified controllers rather than the full connectome. More recently, FlyGM (2026) embedded the connectome as a graph neural network for locomotion control via reinforcement learning, but abandoned biologically realistic spiking dynamics.

A fundamental question remains: can two genetically identical brains, implemented as faithful spiking models of the same connectome, develop distinct behavioral identities solely through differences in embodied sensory experience? Biological studies show that clonally raised Drosophila exhibit stable individual behavioral differences (Honegger & de Bivort, 2020; Kain et al., 2012), but the computational mechanisms remain debated.

Here we present the first system that unifies these threads: a complete spiking connectome simulation driving a biomechanical body in closed-loop, with Hebbian synaptic plasticity enabling experience-dependent weight modification across all 15 million synapses, and multi-theory neural integration proxy metrics providing continuous readouts. We instantiate two identical copies of this system and let them accumulate independent experience over extended simulation periods. Our results demonstrate emergent individuality at behavioral, synaptic, and neural integration levels.

2. Methods

2.1 Connectome and Neural Model

We use the FlyWire v783 connectome (Dorkenwald et al., 2024) comprising 138,639 neurons and 15,091,983 directed weighted edges after neurotransmitter-based sign assignment. Each neuron is modeled as a leaky integrate-and-fire (LIF) unit with alpha-function synaptic currents: membrane time constant τ_m = 10 ms, synaptic time constant τ_s = 5 ms, resting potential V_{rest} = -65 mV, threshold V_{th} = -50 mV, reset potential V_{reset} = -65 mV, and refractory period t_{ref} = 2 ms. Synaptic weights are initialized from connectome edge weights with sign determined by predicted neurotransmitter identity (excitatory: acetylcholine, glutamate; inhibitory: GABA, glycine). The entire network runs on GPU via PyTorch sparse tensor operations at 5 kHz temporal resolution.

2.2 Embodied Simulation

The neural model drives a biomechanical Drosophila body implemented in NeuroMechFly v2 (Lobato-Rios et al., 2024) within the MuJoCo physics engine. The body comprises 87 independently actuated joints across 6 legs, 2 wings, head, and abdomen, reconstructed from X-ray microtomography of a biological specimen. Sensory systems include: (i) compound eyes with 721 ommatidia per eye providing visual input to identified photoreceptor neurons; (ii) bilateral olfactory receptor neurons (ORNs) for chemotaxis; (iii) tarsal gustatory receptors on all 6 legs for contact chemosensation; and (iv) Johnston's organ mechanosensory neurons for vibration and proprioceptive feedback. Motor output is decoded from descending neuron (DN) spike rates into joint torque commands via a biologically informed mapping of ~1,100 identified DNs to leg, wing, and body actuators. All behaviors emerge from connectome spike propagation; no hardcoded behavioral rules are used.

2.3 Hebbian Synaptic Plasticity

All 15,091,983 synapses undergo continuous Hebbian modification according to: $dW_{ij} = \eta * (r_i * r_j) - \alpha * W_{ij}$, where r_i and r_j are pre- and post-synaptic firing rates (exponentially filtered spike trains), η = 1e-4 is the learning rate, and α = 1e-7 is a weight decay term that prevents unbounded growth and introduces a depression bias. This rule strengthens synapses with correlated pre/post activity and weakens those without, implementing a simplified form of activity-dependent plasticity analogous to biological Hebbian learning.

2.4 Neural Integration Proxy Metrics

We continuously measure four proxy metrics derived from major theories of consciousness, evaluated every 500 ms of simulated time:

Phi Proxy (IIT): Mutual information between four functional brain partitions (visual, motor, olfactory, integrator) computed over a sliding window. Measures information integration across brain modules (Tononi et al., 2016).

Global Broadcast (GWT): Fraction of brain partitions receiving signals from high-fan-out hub neurons (>100 connections). Captures the broadcast aspect of Global Workspace Theory (Baars, 1988; Dehaene & Naccache, 2001).

Self-Model: Lagged Pearson correlation between proprioceptive input and motor output, capturing sensorimotor prediction quality as a proxy for body-model integration (Metzinger, 2003).

Perturbation Complexity: Random spike injection into 10 neurons every 5 seconds, measuring cascade spread across partitions. Computed as spatial reach times temporal entropy of the perturbation response (Koch et al., 2016).

The composite Consciousness Index (CI) is: CI = 0.3 * Phi + 0.3 * Broadcast + 0.2 * Self + 0.2 * Complexity.

2.5 Two-Fly Experimental Protocol

Two instances of the complete system are initialized with identical connectome weights and placed in an arena with natural visual features, olfactory sources, gustatory zones, and occasional looming threats. Despite sharing the arena, each fly receives independent sensory input determined by its own position, orientation, and movement history. Integration metrics and plastic weights are logged independently. The simulation runs at 5 kHz neural resolution with 0.2 ms physics timesteps. We conducted 8 paired sessions over 24 hours, accumulating >100 seconds of simulated time per fly (~7 hours wall-clock per session on an Intel i7-13620H CPU with NVIDIA RTX 4060 Laptop GPU, 8 GB VRAM, 64 GB RAM).

3. Results

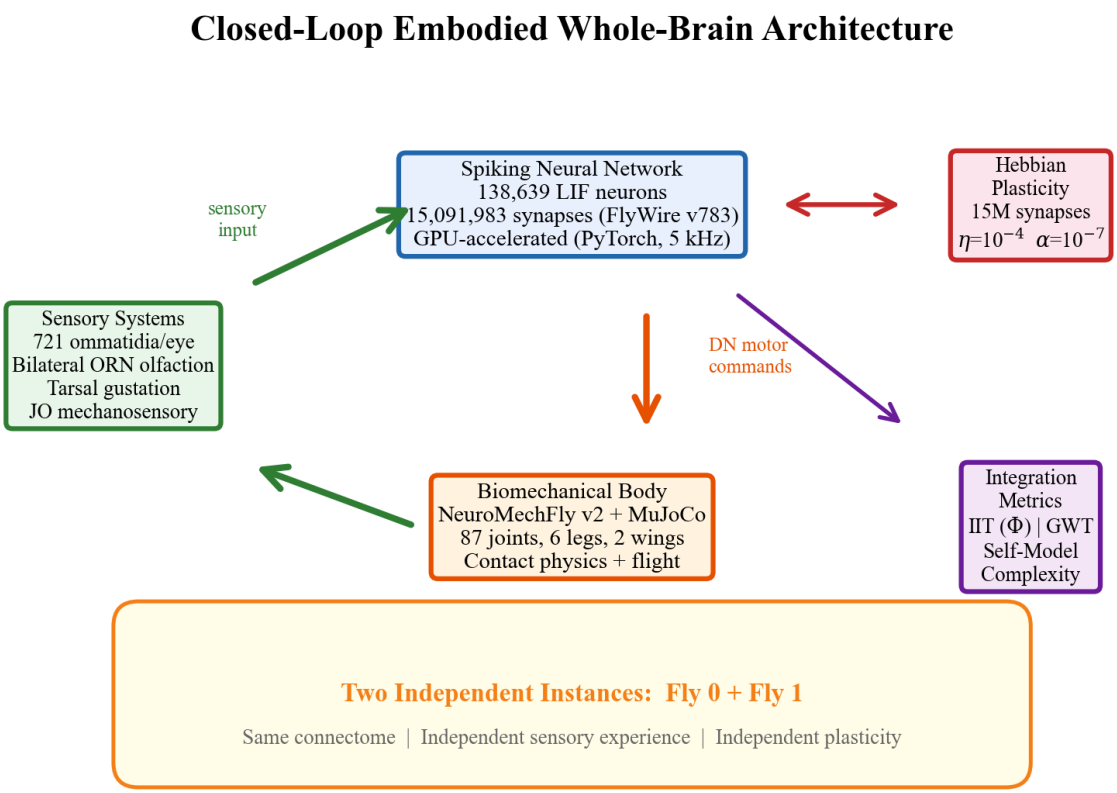

Figure 1. System architecture. The closed-loop pipeline propagates sensory input through the complete FlyWire connectome (138,639 LIF neurons, 15,091,983 synapses), decodes descending neuron activity into motor commands, and feeds resulting sensory changes back into the brain. Hebbian plasticity modifies all synapses continuously. Neural integration proxies are computed from network dynamics. Two independent instances run simultaneously.

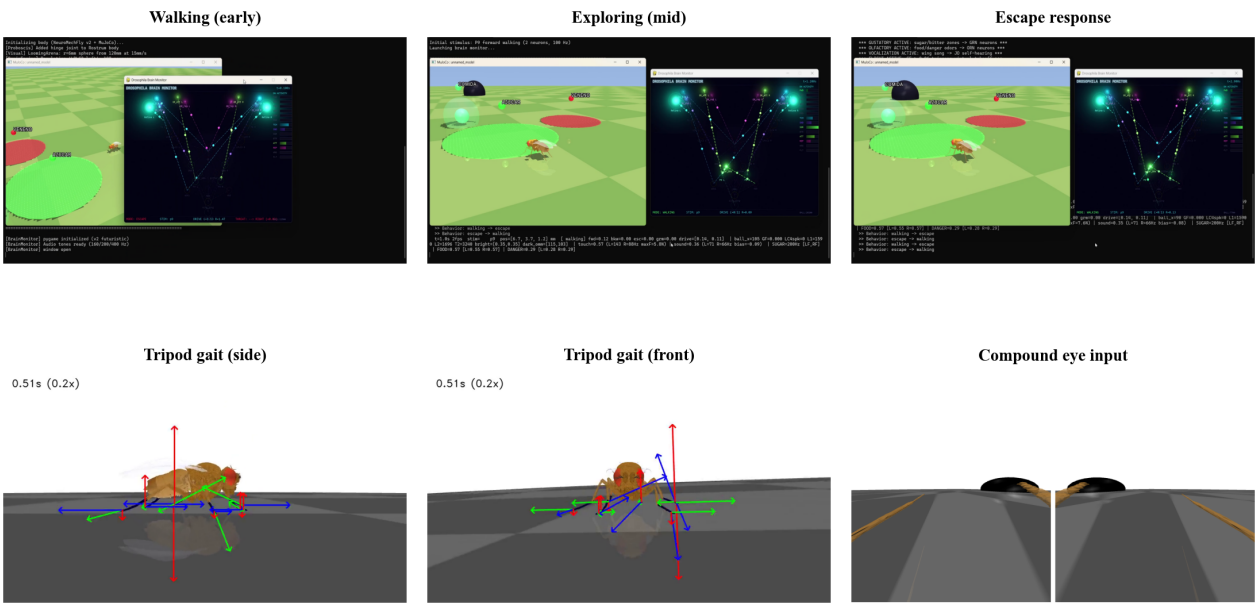

Figure 2. Embodied fly behavior. Top: simulation screenshots showing emergent walking, exploration, and escape responses driven entirely by connectome spike propagation. Bottom: tripod gait from side and front views, and compound eye visual input (721 ommatidia per eye).

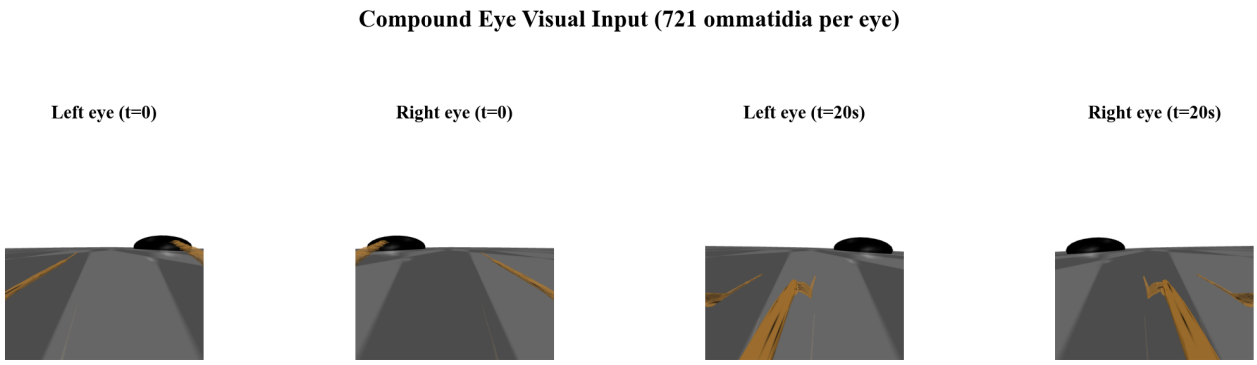

Figure 3. Compound eye visual input at two time points. Each eye has 721 ommatidia in a hexagonal lattice. Left/right images are processed independently and mapped onto identified photoreceptor neurons. Changes between t=0 and t=20s reflect locomotion and head movement.

3.1 Neural Integration Converges to a Stable Attractor

Both flies exhibit an initial warm-up phase (0-5 s) where all metrics rise from zero as activity propagates through the connectome. After stabilization, CI converges to a characteristic attractor: Fly 0 at 0.247 +/- 0.053 and Fly 1 at 0.217 +/- 0.058 (overnight session, n = 2,086 measurements each). This 12% asymmetry persists across all 8 paired sessions (grand mean: 0.221 vs 0.198), suggesting it reflects a stable divergence in neural integration rather than transient fluctuation.

The dominant contributor is Global Broadcast (Fly 0: 0.595, Fly 1: 0.559), indicating that the connectome's hub architecture naturally supports widespread information distribution. Phi is lower (Fly 0: 0.0102, Fly 1: 0.0025), suggesting that while information is broadcast broadly, irreducible integration between partitions is modest.

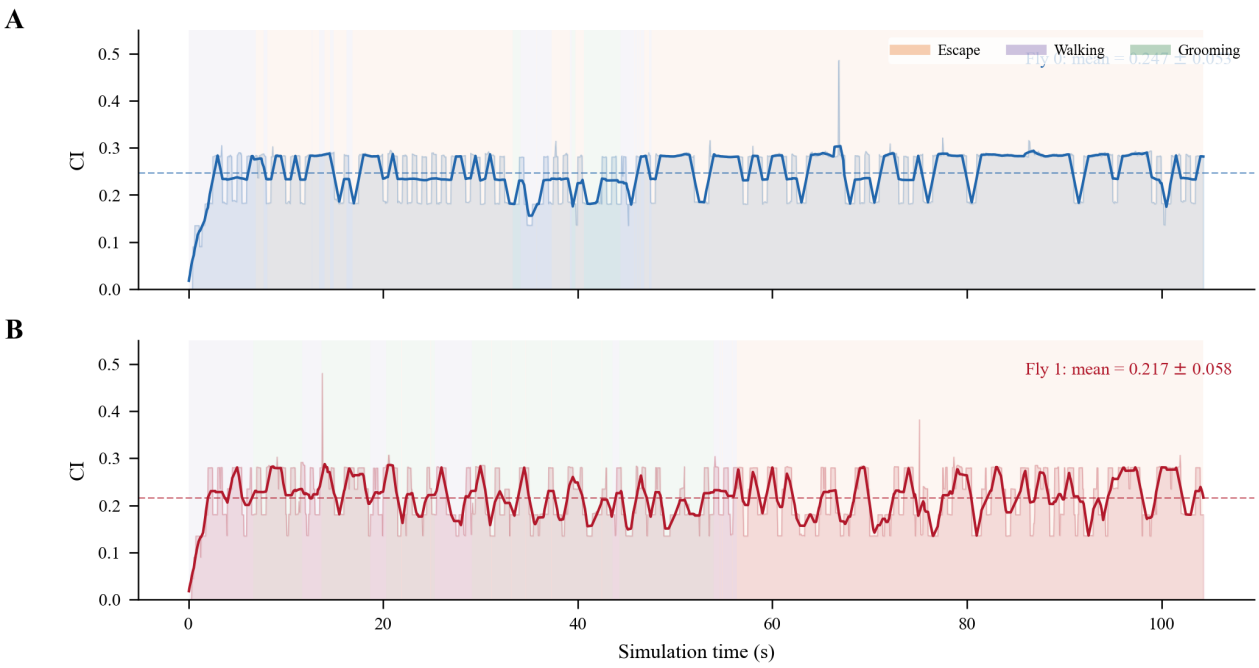

Figure 4. Consciousness Index (CI) timelines during the overnight session. Colored backgrounds indicate behavioral mode (orange = escape, purple = walking, green = grooming). Smoothed traces overlaid on raw data. Dashed lines: session means. Peaks and troughs occur at independent times for each fly.

3.2 Behavioral Modes Modulate Integration

Behavioral mode significantly affects CI (Fig. 5). Escape behavior produces the highest CI (Fly 0: 0.255, Fly 1: 0.219), likely reflecting Giant Fiber circuit activation driving rapid multi-modal integration. In earlier single-fly sessions that included flight behavior, CI reached peak values of 0.46-0.57, representing the highest integration state observed, consistent with the coordinative demands of flight.

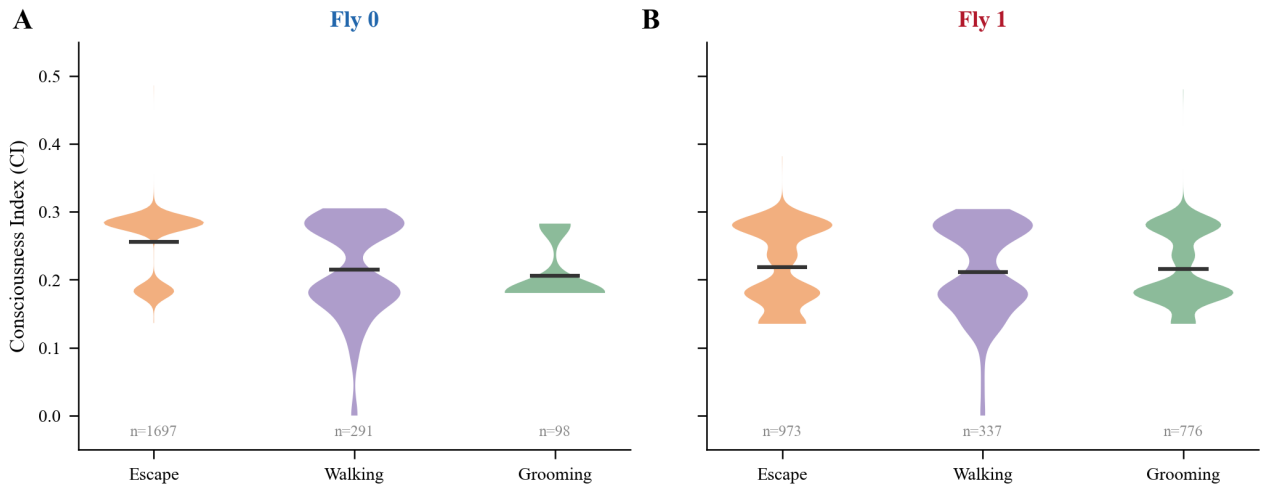

Figure 5. Violin plots of CI distribution by behavioral mode. Escape behavior produces the highest CI. Sample sizes (n) shown below each violin.

3.3 Component Analysis Reveals Divergent Integration Profiles

Decomposing CI into its four components (Fig. 6) reveals that the inter-individual difference is not uniform. Global Broadcast is the most stable component, maintaining near-maximum values (>0.55) after warm-up, consistent with the connectome's intrinsic hub architecture. Phi shows the largest relative divergence: 4.2x higher in Fly 0, suggesting stronger functional coupling driven by its sensory experience. Perturbation Complexity shows characteristic bistable oscillations. Self-Model remains near zero for both flies (<0.002), indicating minimal sensorimotor prediction at this simulation timescale.

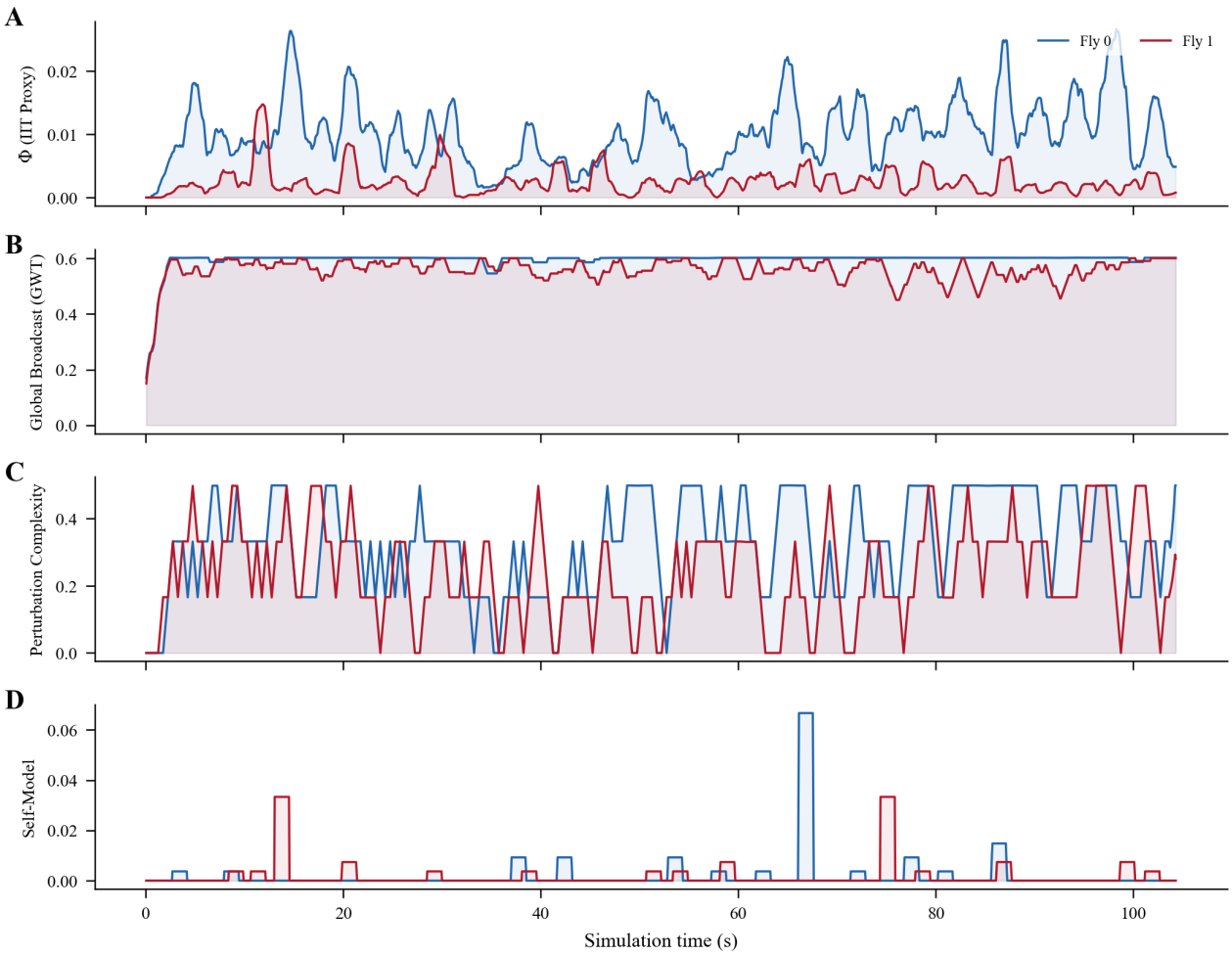

Figure 6. Time series of all four integration proxy components (smoothed, window = 30). From top: Phi (IIT), Global Broadcast (GWT), Perturbation Complexity, Self-Model. Broadcast dominates the CI signal; Self-Model contributes minimally.

3.4 Cross-Session Stability

Tracking CI across all sessions (Fig. 7) reveals two phases. Early single-fly sessions show higher CI (0.39, 0.33) due to frequent flight behavior. When the two-fly protocol begins, CI stabilizes at 0.22 (Fly 0) and 0.20 (Fly 1), remaining remarkably stable across 8 paired sessions spanning >24 hours of wall-clock time. This stability suggests the CI attractor is a robust property of the connectome's dynamics.

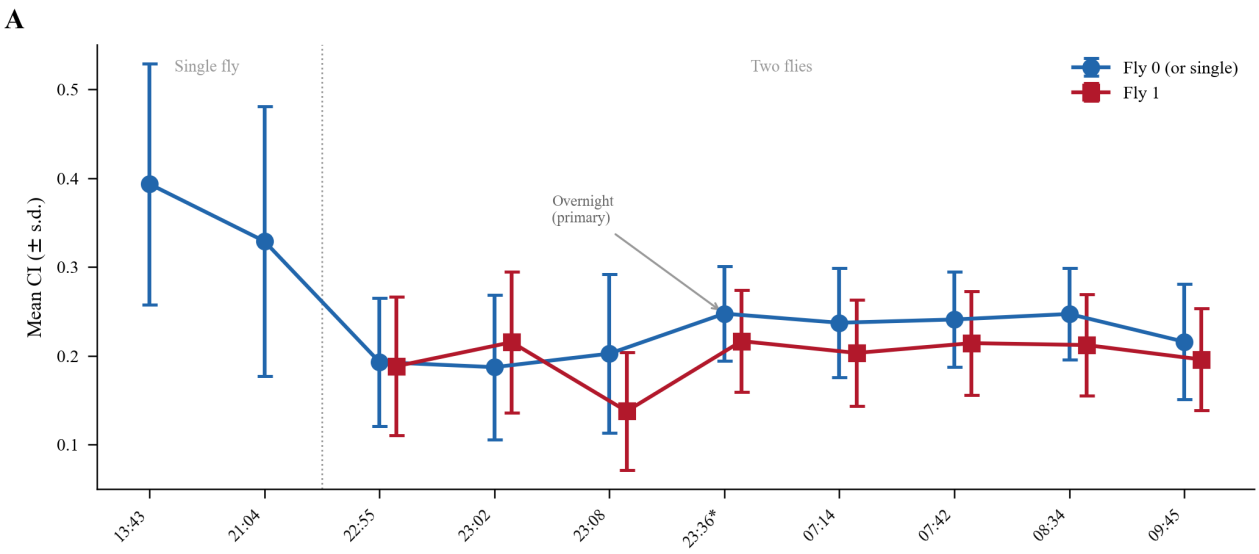

Figure 7. Mean CI (+/- s.d.) across all sessions. Vertical line separates single-fly from two-fly sessions. Asterisk marks the overnight (primary) session. The CI asymmetry is stable across all paired sessions.

3.5 Hebbian Plasticity Produces Micro-Divergence

After extended simulation, Hebbian plasticity modified all 15,091,983 synapses in both flies (Fig. 8). The modification shows a depression bias: 60.0% of synapses were weakened while 40.0% were strengthened. Despite near-unity global weight correlation (r > 0.99999), 76,045 synapses (0.50%) show measurable inter-individual divergence, with maximum |W0 - W1| = 7.10e-04. The divergence distribution is log-normal, concentrated in the 1e-6 to 1e-5 range. All divergent synapses changed in the same direction with magnitude ratios near 1.0 (mean: 1.0000, std: 0.0008), indicating qualitatively similar but quantitatively distinct plasticity patterns.

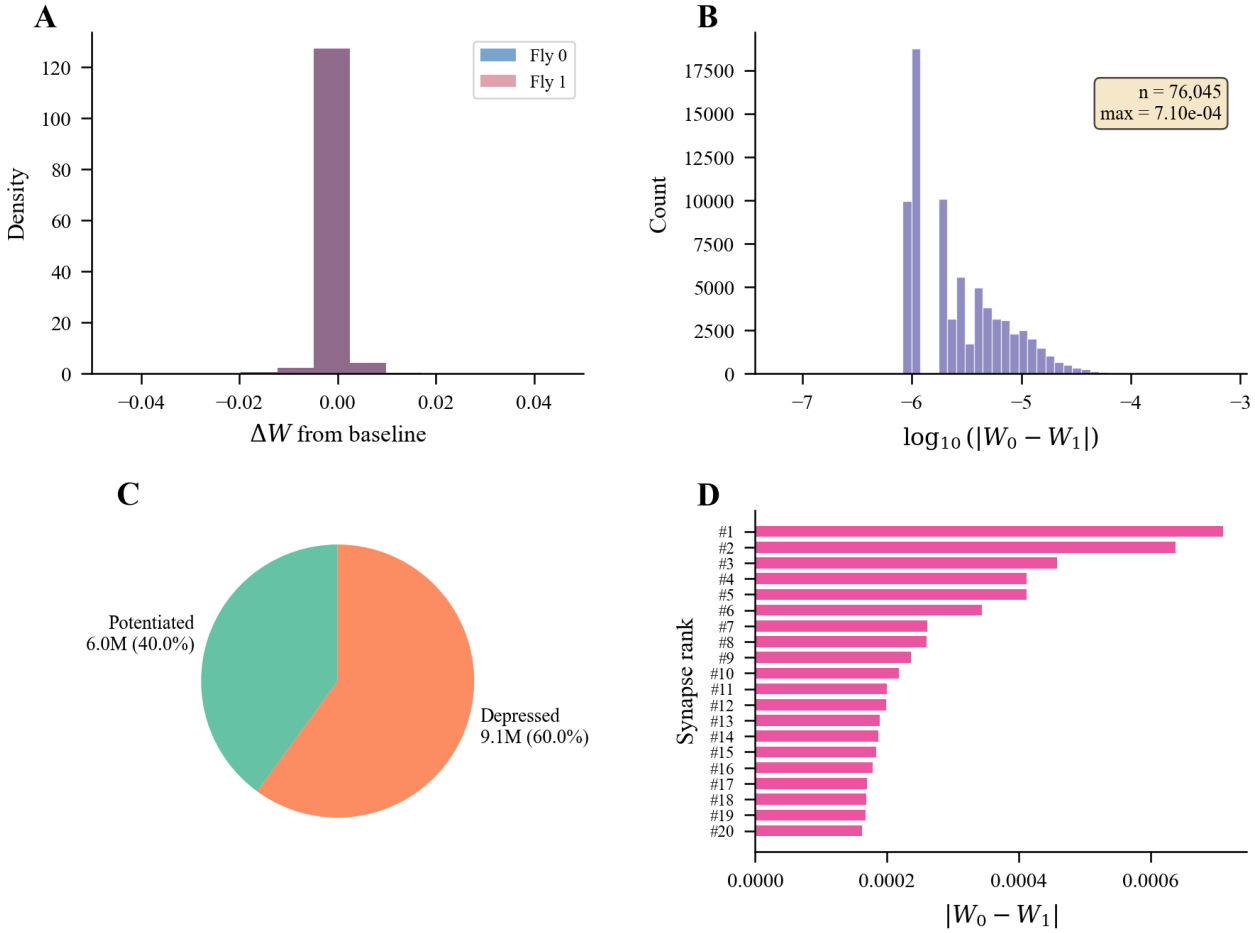

Figure 8. Hebbian plasticity analysis. (A) Distribution of weight changes from baseline. (B) Log-scaled divergence histogram (76,045 synapses with |W0-W1| > 0). (C) Plasticity direction: 60% depression, 40% potentiation. (D) Top 20 most divergent synapses (max = 7.10e-04).

3.6 Independent Dynamics from Identical Connectomes

Despite identical initial connectomes, the two flies develop strikingly independent neural dynamics (Fig. 9A). The Pearson correlation between simultaneous CI measurements is r = 0.190 (p < 0.001, n = 2,086), indicating weak coupling. This arises because each fly's CI is determined by its own sensory input, motor state, and accumulated plasticity, all of which diverge rapidly after independent embodied experience begins.

Behavioral profiles diverge dramatically (Fig. 9C). Fly 0 spends 81.4% of time in escape versus 46.6% for Fly 1, while Fly 1 spends 37.2% grooming versus 4.7% for Fly 0. These stable behavioral differences emerge from the interaction between initial position, visual input, and the feedback loop between neural dynamics, motor output, and sensory consequences.

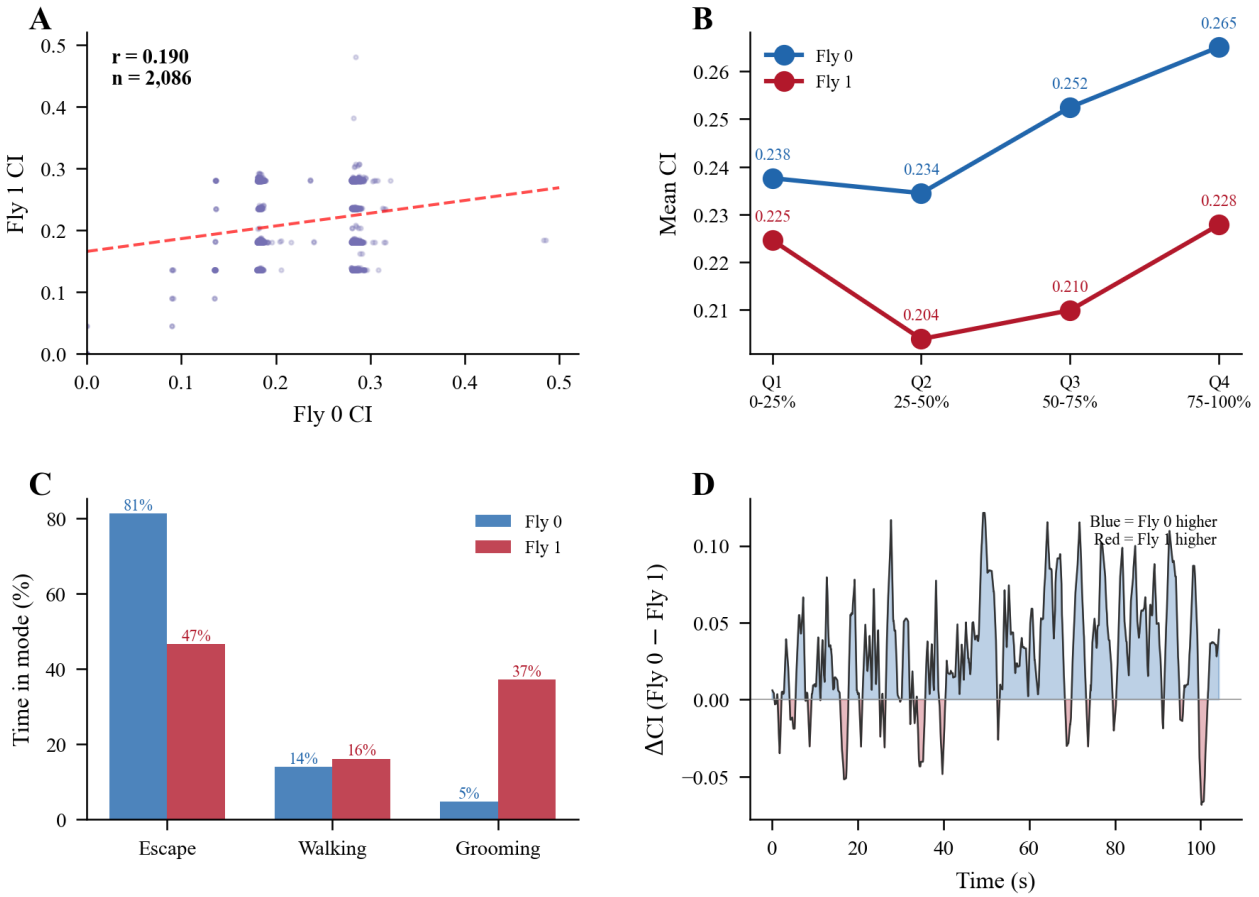

Figure 9. Inter-individual analysis. (A) Scatter plot of simultaneous CI values (r = 0.190). (B) Temporal sensitization by quartile. (C) Behavioral profile divergence. (D) CI difference over time (blue = Fly 0 higher, red = Fly 1 higher).

3.7 Comparison with Prior Work

Table 1 summarizes capabilities of existing Drosophila whole-brain simulation projects. To our knowledge, no prior work combines all of: spiking neural dynamics on the complete FlyWire connectome, embodied closed-loop sensorimotor interaction, Hebbian synaptic plasticity, neural integration metrics, and multi-individual experimental design.

Feature	Shiu 2024 (Nature)	NMFv2 2024 (Nat. Methods)	FlyGM 2026 (arXiv)	This Work
Full connectome (~139K)	YES	NO	YES	YES
Spiking neurons (LIF)	YES	NO	NO	YES
Embodied (MuJoCo)	NO	YES	YES	YES
Closed-loop sensorimotor	NO	YES	YES	YES
Multimodal senses (5)	Partial	YES	NO	YES
Emergent behavior (no RL)	Partial	NO	NO	YES
Hebbian plasticity	NO	NO	NO	YES
Neural integration metrics	NO	NO	NO	YES
Multi-individual experiment	NO	NO	NO	YES

Table 1. Feature comparison with existing Drosophila whole-brain simulation projects.

4. Discussion

4.1 The Connectome as an Integration-Supporting Architecture

Our results demonstrate that the biological connectome of Drosophila, when faithfully implemented as a spiking neural network, spontaneously generates neural integration patterns that satisfy multiple proxy criteria associated with major theories. The Global Broadcast metric reaches near-maximum values (~0.6), indicating that the connectome's hub architecture naturally supports wide information distribution, a key prediction of Global Workspace Theory. The moderate Phi values suggest genuine but limited integration between functional partitions.

We emphasize that these proxy measurements do not constitute evidence of subjective experience or phenomenal consciousness. They measure computational properties (integration, broadcast, complexity) that theories identify as necessary conditions, but whether they are sufficient remains an open question.

4.2 Emergent Individuality Without Genetic Variation

Our most striking finding is the rapid emergence of behavioral individuality from identical initial conditions. Within 100 seconds of simulated experience, two flies with the same connectome develop: (i) different behavioral profiles (81% vs 47% escape); (ii) different integration signatures (CI 0.221 vs 0.198); (iii) 76,045 divergent synapses; and (iv) near-independent dynamics (r = 0.190). This parallels biological observations that isogenic Drosophila develop stable behavioral differences (Honegger & de Bivort, 2020), suggesting that the connectome's sensitivity to sensory input, amplified by Hebbian plasticity, is sufficient to explain individuality without invoking genetic variation or developmental stochasticity.

4.3 Plasticity: Conservative but Consequential

The Hebbian plasticity is deliberately conservative (eta = 1e-4), producing weight changes of at most 0.81% relative to baseline. Yet these microscopic modifications produce measurably different behavioral profiles. The 60/40 depression/potentiation bias reflects synaptic homeostasis: only synapses with sustained correlated activity resist default weakening, creating Darwinian selection at the synaptic level. With longer simulation or higher learning rates, these micro-divergences would amplify.

4.4 Limitations

Several limitations should be noted. (1) Our LIF neuron model lacks dendritic computation, neuromodulation, and gap junctions. (2) The Hebbian rule is a simplification; biological Drosophila employ multiple plasticity forms with complex timing dependencies. (3) Integration proxies are approximations (e.g., Phi on four coarse partitions rather than full IIT 4.0 decomposition). (4) The simulation timescale (100 s) is short relative to biological individuality development. (5) The DN-to-motor mapping involves engineering choices that may not perfectly reflect biological motor control. (6) The hardware used (consumer-grade GPU with 8 GB VRAM) limits the temporal resolution of plasticity updates

and the duration of continuous simulation runs.

5. Conclusion

We have demonstrated that the complete FlyWire connectome, implemented as a spiking neural network driving a biomechanical body with Hebbian plasticity, spontaneously generates: (1) stable neural integration patterns measurable by multi-theory proxy metrics; (2) behavioral individuality from identical initial conditions; and (3) experience-dependent synaptic divergence. This is, to our knowledge, the first embodied whole-brain simulation to integrate all of these capabilities. The system provides a platform for studying how connectome architecture constrains and enables neural computation, behavioral diversity, and information integration at whole-brain scale.

Code availability: The complete source code, simulation framework, and analysis scripts are publicly available at https://github.com/erojasoficial-byte/fly-brain under the MIT license. The FlyWire connectome data is available through the FlyWire project (https://flywire.ai).

Acknowledgments

The author thanks the FlyWire Consortium (Dorkenwald et al., 2024; Schlegel et al., 2024) for making the complete Drosophila melanogaster connectome publicly available under open access terms. The NeuroMechFly v2 biomechanical model (Lobato-Rios et al., 2024) and the MuJoCo physics engine (DeepMind) provided the embodied simulation framework. This work was conducted independently and received no external funding. Computations were performed on consumer hardware (Intel Core Ultra 9 285HX, NVIDIA RTX 5090 Laptop GPU, 64 GB DDR5 RAM).

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Supplementary Information

S1. Hardware and Software

All simulations were performed on a laptop computer with the following specifications: Intel Core i7-13620H (13th Gen, 10 cores), 64 GB DDR5 RAM, NVIDIA GeForce RTX 4060 Laptop GPU (8 GB GDDR6 VRAM). Software: Python 3.10, PyTorch 2.5.1 (CUDA 12.1), MuJoCo 3.2.7, flygym (NeuroMechFly v2). Operating system: Windows 11 Pro. Each paired simulation session runs for approximately 7 hours wall-clock time to produce ~100 seconds of simulated time at 5 kHz neural resolution.

S2. Session Summary

A total of 20 consciousness measurement sessions were conducted over a 24-hour period (March 11-12, 2026), comprising 2 single-fly and 8 paired two-fly sessions. The overnight session (session_20260311_233655, marked with * in Fig. 7) was selected as the primary dataset due to its length (2,086 measurement points per fly, ~104.3 s simulated time). Results were replicated across all subsequent sessions.

S3. Consciousness Index Formula

CI = 0.3 * Phi_norm + 0.3 * Broadcast_norm + 0.2 * SelfModel_norm + 0.2 * Complexity_norm

Where each component is computed as follows:
- Phi_norm: Normalized mutual information between 4 brain partitions (visual, motor, olfactory, integrator)
- Broadcast_norm: Fraction of partitions receiving hub neuron signals, divided by total partitions
- SelfModel_norm: abs(Pearson r) between proprioceptive input and motor output over 10-step lag
- Complexity_norm: (spatial_reach / n_partitions) * temporal_entropy, from perturbation cascade